

SmartCare v0.2 - Requirements Specification Template

1. Problem and Scope

SmartCare currently has several operational problems due to workplace practices where records are made both digitally and manually, primarily causing:

- duplicate appointment bookings and patient records;
- difficulties locating patient records;
- inconsistent appointment statuses, further caused by manual cancellation processes;
- difficulties finding practitioner availability;
- an unreliable appointment history;
- and difficulties producing operational reports.

As a result, the client requests a simple software system that can improve patient, practitioner and appointment management.

In Scope	Out of Scope	Provisional
• Patient registration • Appointment management (creation / cancellations, statuses) • Cancelled appointments remain in history • Prevent of double-booking practitioners • Prevention of duplicate patient records • Patient search • Produce operational reports • Save data during sessions	• Facial recognition login • Online payment • AI-recommended treatments • AI diagnosis recommendations • Insurance processing	• Patient SMS reminders • Appointment management permissions • Practitioner schedule view

2. Stakeholders

Stakeholder	Need	Evidence
Patient	Book appointments and have accurate patient information and appointment details	Addresses issue regarding "inconsistent appointment status information"
Practitioner	View patient and appointment information and have a clear schedule where patient appointments do not overlap	Addresses the issue regarding "limited visibility of practitioner availability"
Receptionist	Record and patient information, create and cancel appointments without creating data redundancy	Addresses the issue regarding "lack of reliable appointment history"

Clinic Manager	A simple software system that manages clinic operations and ensure patient, staff and appointment information are correct	Addresses the core goal: "Management wants a simple software system that can initially support patient, practitioner and appointment management."
System Administrator	A system than can be maintained, tested and conserve consistent data	Addresses the requirement: "The first version should be a manageable application suitable for a small clinic."

3. Functional Requirements

FR-01: The system must allow staff to register patients

FR-02: The system must allow staff to register practitioners

FR-03: The system must allow practitioners to view patient information and appointment details

FR-04: The system must allow receptionists to view practitioner availability

FR-05: The system must allow staff to create appointments

FR-06: The system must allow staff to cancel appointments

FR-07: The system must prevent staff from allocating appointments at the same time for one practitioner

FR-08: The system must prevent staff from inputting invalid patient/appointment information

FR-09: The system must maintain relevant patient, appointment and staff data during sessions

FR-10: The system must allow staff to search for patients in the system

FR-11: The system must generate operational reports

FR-12: The system must maintain appointment status information for every patient

4. Non-Functional Requirements

NFR-01: The system must ensure recorded data (patient information, appointment details, etc.) remains in the system even if it the system has shut down

NFR-02: The system must handle invalid inputs or unexpected errors whilst ensuring said issues never causes the system to crash

NFR-03: The system must be designed in separated components to allow the possibility of new features to be developed in the future without breaking existing functions.

NFR-04: The system must provide clear and easy-to-understand messages when an input is successful or when an error occurs.

NFR-05: The system must ensure that any updates to data such as appointment statuses are instantly updated across all schedules to prevent inconsistent information

NFR-06: The system must allow individual functions to be tested separately without running the whole system

5. User Stories

US-01: As a receptionist, I want to register a patient into the system, so that patient information is accessible and recorded for future visits.

US-02: As a practitioner, I want to set my weekly availability time, so that I am not allocated two appointments at the same time for two different patients.

US-03: As a receptionist, I want to search for a patient on the system, so that I can efficiently find the relevant patient information.

US-04: As a receptionist, I want to create appointments, so that patients are allocated to a practitioner.

US-05: As a clinic manager, I want to view clinic operations and data, so that I can monitor clinic performance and appointment operations.

US-06: As a receptionist, I want to cancel appointments, so that practitioner availability is freed up for other patients while keeping appointment information/statuses accurate.

6. Acceptance Criteria

Patient Registration Acceptance Criteria

GIVEN: the receptionist is registering a patient into the system

WHEN: the receptionist enters the patient's details (e.g. first name, last name, age, etc.)

THEN: the system adds the information into the list of registered patients and allows staff to view patient details prior to future visits

Patient Search Acceptance Criteria

GIVEN: the receptionist is searching for an existing patient in the system

WHEN: the receptionist submits a search query matching a patient's name

THEN: the system returns all patient records that matches the query string

Negative Acceptance Criteria

GIVEN: the practitioner already has an appointment with a patient at a specified time

WHEN: staff registers another patient with the occupied practitioner at the same specified time

THEN: the system rejects the input and prints an error statement and warning staff that the practitioner is not available

7. Assumptions and Open Questions

Patient cancellations are kept in the system

At this stage, we can assume that when an appointment cancellation occurs, the record of the cancellation is kept within the system regardless of whether they were not present for consultation. This assumption is primarily made

based on retention laws that require businesses to hold patient records for a minimum of a few years before they can dispose of unnecessary data. In addition, it could serve as legal or financial evidence for the clinic.

OQ: Will time slots of cancelled appointments be available to book again?

Patient search is a confirmed requirement

During stage 1, it was stated that patient search is a provisional feature as it was not explicitly requested by the client. In addition, the AI suggestion for a search functionality was rejected due to the complexity of implementing the feature primarily based on current coding knowledge and skills.

However, during stage 2 of repairing ambiguous requirements, there is a scenario where: "Patient search should be fast". This could imply that patient search is now a requirement hence this assumption is made.

OQ: Will search results appear whilst staff is entering letters one by one (real-time searching), or will they have to press a button to confirm a search query?

One practitioner can not have two appointments at the same time

In stage 1, a question for the client addresses whether a practitioner can have two appointments allocated to them at the same time. Henceforth, it is now assumed that one practitioner can not have two appointments at the same time as it is physically impossible to manage two different patients simultaneously without causing confusion. In addition, it should alleviate practitioner workload and further demonstrate practical workplaces practices.

OQ: Will keeping cancellation records (as per assumption 1) clash with practitioner availability times? I.e. A practitioner is free due to a cancellation, but will validation errors occur if another patient tries to book at that time?

8. AI Requirements Review Record

AI suggestion	Evidence?	Decision	Reason	Verification
Confirm that the system must prevent duplicate appointment bookings for a practitioner.	FR-07 Case study: "Duplicate appointment bookings"	Accepted	FR-07 addresses a current issue in the case study regarding "duplicate appointment bookings."	Verified Supported via case study and FR-07
Clarify practitioner access to patient information	FR-03	Modified	FR-03 does not define what patient information includes	Partially verified Confirms patient information is needed, but does not specify the required details. Therefore, clarification by the client is required
Define appointment status values	FR-12 Case study:	Modified	FR-12 does not specify appointment status values	Partially verified Confirms appointment status

	"Inconsistent appointment status information"			details are required by the client, but status types are not defined. Therefore, clarification by the client is required
Confirm the handling of cancelled appointments	FR-06	Unverified	Case study does not specify how cancelled appointments are handled	Not verified Neither case study or requirements specify how cancelled appointments are to be addressed. Therefore, confirmation by the client is required
Confirm whether operational reports should be defined more precisely before implementation	FR-11 Case study: "Difficulty producing basic operational reports"	Modified	FR-11 states the system must generate "operational reports", however the report contents are not specified	Partially verified Confirms operational reports are required via client needs, but details regarding the report contents are not specified. Therefore clarification by the client is required.